

RAFAT HOSSAIN

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EDUCATION

Bachelor of Computer Science (Honours), Dalhousie University (CGPA: 3.98/4.30)

Class of 2027

Relevant Courses: Software Development, Data Structures and Algorithms, Database Systems, Operating Systems, Machine Learning, Theory of Computer Science

Activities: Dalhousie Machine Learning Society (Treasurer), Sexton Scholar (top academic standing), Dean's List, 1st Place — Machine Learning, SURGE NeuroHack 2026

EXPERIENCE

Software Developer Co-op, SpryPoint

Remote | **2026**

- Two co-op terms on **SpryCIS**, an enterprise billing and customer information platform for municipal utilities; **returned for a second term by invitation**.
- Delivered **~20 client-facing stories** on a **Scala/Java** monolith built on the **Play Framework** with **Ebean ORM** over **PostgreSQL**, and a **Knockout.js** MVVM front end — bug fixes, new configuration pages and forms, and interface work.
- Owned a **5-point fix on the billing page**, the platform's most sensitive surface, spending most of the estimate establishing what the change could affect before altering anything.
- Wrote unit and UI tests and authored **regression and exploratory test cases** run team-wide at each sprint close; reviewed teammates' pull requests behind **CI** gates on builds, test suites and commit conventions.

Undergraduate Research Assistant, Dalhousie University

Halifax, NS | **2025 – Present**

- Built and trained **U-Net** segmentation models (**Python, PyTorch**) on a Landsat-8 archive over West Greenland, and built the patch datasets four other researchers on the team train against.
- Validated output against **independently published ice masks** rather than the pseudo-labels the model learned from — establishing that the easier evaluation was circular and would have overstated the result.
- Traced a season-mislabelled dataset export that had silently mixed summer and winter imagery across the team's training runs.

Treasurer, Dalhousie Machine Learning Society

Halifax, NS | **2025 – Present**

- Secured **\$1,500+ in grant funding** across two terms for a society of **150+ members**, authoring successful applications to the **Dalhousie Student Union** and **ShiftKey Labs**.
- Own budgeting, invoicing and reconciliation across the academic year; **DSU Treasurer Training** certified.

Data Architect, Knownautoparts

Remote | **2023 – 2025**

- Designed a relational product database for **1,000+ SKUs** with data-quality controls that kept the catalogue accurate, and restructured product taxonomy for search — increasing organic traffic **35%**.

PROJECTS

Canon Film Scanners — Desktop Applications

2026

- Two **Python** applications controlling a Canon EOS R7 over USB and Wi-Fi with live image processing and remote capture; CI on both repositories.
- Benchmarked both transports (3.98 vs 59.79 fps), diagnosed a protocol limit with no USB path, and ported to a second SDK rather than optimise a dead end — **15× throughput**.

SPECTRA-ICA — EEG Artifact Removal (1st Place, ML Category, SURGE NeuroHack 2026)

2026

- Python, MNE, scikit-learn**. Removes only artifact-characteristic frequencies during detected artifact windows instead of discarding whole independent components.
- Beat standard ICA on **all 7 metrics** in a P300 benchmark under leave-one-person-out cross-validation across 10 subjects.

CERTIFICATIONS

AWS Certified Cloud Practitioner

Amazon Web Services

Fundamentals of Cloud Administration with Azure

ShiftKey Labs

Practical Cybersecurity

ShiftKey Labs

Level Up with CCubed — Cybersecurity

Computing Councils of Canada